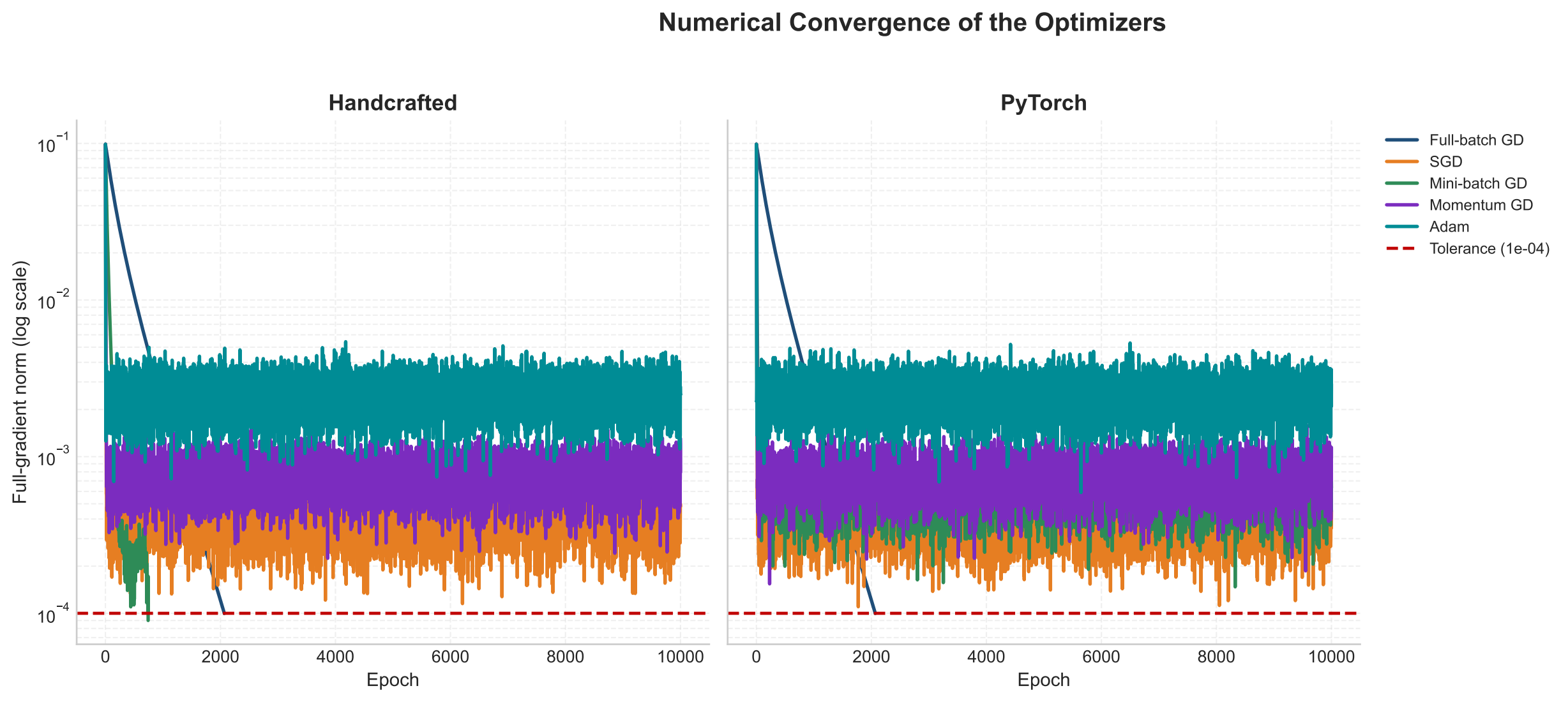

Numerical Convergence of the Optimizers
Handcrafted
PyTorch
Full-gradient norm (log scale)
10^{-1}
10^{-2}
10^{-3}
10^{-4}
0
2000
4000
6000
8000
10000
Epoch
Full-batch GD
SGD
Mini-batch GD
Momentum GD
Adam
Tolerance (1e-04)